Exploratory Data Analysys

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This code uses static data as its starting point rather than running a lengthy API call. For production, to access realtime current data, this would be replaced by a call to the API using the reticulate functions that call Python code in R

```
RawData <- readRDS(here::here("Data","PreEDA_DataFrame.rds"))
names(RawData) <- make.unique(names(RawData), sep = ".")
FieldActions <- read_excel(here::here("Data", "FieldActions.xlsx"))
CleanedData <- RawData # make a copy to preserve original
IncomeData <- read_excel(here::here("Data", "Income Group Data.xlsx")) # Import income data
```

```
# This code iterates through every field in the raw data, taking the cleanup action on
# it that is specified in the FieldActions data frame.  The net result is that a bunch of
# duplicated and/or unneeded fields are deleted, fields with uninterpretable (coded) column
# names tied to their source are renamed to be descriptive, and the Year column is forced
# to an integer

for (fieldcounter in names(CleanedData)) {
  action_row <- FieldActions[FieldActions$FieldName == fieldcounter, ]
  if (nrow(action_row) == 0) {
    next  # No action specified for this field
  }
  WhatToDo <- action_row$Action
  if (WhatToDo == "Delete") {
    CleanedData[[fieldcounter]] <- NULL

  } else if (WhatToDo == "Rename") {
    names(CleanedData)[names(CleanedData) == fieldcounter] <- action_row$RenamedTo

  } else if (WhatToDo == "Make Integer") {
```

```r
    CleanedData[[fieldcounter]] <- as.integer(CleanedData[[fieldcounter]])
  }
}
# Now, pull in the Income Group Data, and join it to the raw data by CountryCode
CleanedData <- left_join(CleanedData, IncomeData, by = c("CountryCode" = "Country Code"))

# Now for initial EDA purposes, I am only going to consider the two wellbeing indexes
# which are tagged as my PrimaryResponse fields.  So I want to drop the other more granular
# Response fields

# Select field names to keep
# First, all the renamed fields that weren't dropped and aren't a Response
PrimaryFields <- FieldActions %>%
  filter(Type %in% c("PrimaryResponse","Predictor","Identifier"),
         is.na(Action) | Action != "Delete") %>%
  pull(RenamedTo)
# Now add the Income Group field (since that was joined from a different data set)
PrimaryFields <- c(PrimaryFields, "Income Group")
# Subset the cleaned dataset using that list of fields
FirstCutData <- CleanedData %>%
  select(all_of(PrimaryFields))
write_xlsx(FirstCutData, path = here::here("Data","FirstCutData.xlsx"))
```

Table 1: Summary Statistics (Variables 1 to 8)

CountryCode	CountryName	year	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore
Length:5974	Length:5974	Min. :1995	Min. :0.2380	Min. :0.1340	Min. : 4.819	Min. : 17.00	Min. :-1.96956
Class :character	Class :character	1st Qu.:2002	1st Qu.:0.5700	1st Qu.:0.3990	1st Qu.:11.060	1st Qu.: 76.58	1st Qu.:-0.82041
Mode :character	Mode :character	Median :2009	Median :0.7110	Median :0.5930	Median :19.183	Median : 94.52	Median :-0.29892
NA	NA	Mean :2009	Mean :0.6899	Mean :0.5783	Mean :21.317	Mean : 91.33	Mean :-0.05974
NA	NA	3rd Qu.:2016	3rd Qu.:0.8140	3rd Qu.:0.7500	3rd Qu.:31.878	3rd Qu.:102.80	3rd Qu.: 0.59246
NA	NA	Max. :2023	Max. :0.9720	Max. :0.9230	Max. :51.891	Max. :578.71	Max. : 2.45912
NA	NA	NA	NA's :422	NA's :3677	NA's :3670	NA's :720	NA's :1204

Table 2: Summary Statistics (Variables 9 to 16)

ElectricAccess	PopDensity	PopInSlums	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex
Min. : 0.80	Min. : 1.438	Min. : 0.00	Min. : 0.0258	Min. : 0.40	Min. :0.0054	Min. :-2.44023	Min. : 0.6032
1st Qu.: 63.20	1st Qu.: 30.875	1st Qu.: 1.55	1st Qu.: 3.7148	1st Qu.: 31.07	1st Qu.:0.2383	1st Qu.:-0.78651	1st Qu.: 7.5392
Median : 98.10	Median : 77.922	Median :21.18	Median : 11.1751	Median : 54.05	Median :0.6029	Median :-0.20432	Median : 15.1677
Mean : 79.23	Mean : 298.200	Mean :28.73	Mean : 65.0840	Mean : 57.04	Mean :0.9675	Mean :-0.06367	Mean : 25.1422
3rd Qu.:100.00	3rd Qu.: 176.610	3rd Qu.:52.80	3rd Qu.: 39.6974	3rd Qu.: 86.76	3rd Qu.:1.3904	3rd Qu.: 0.59433	3rd Qu.: 34.1564
Max. :100.00	Max. :18822.891	Max. :99.81	Max. :3850.5000	Max. :100.00	Max. :6.0192	Max. : 2.46966	Max. :171.1775
NA's :627	NA's :583	NA's :4011	NA's :1505	NA's :5412	NA's :3666	NA's :1223	NA's :3592

Table 3: Summary Statistics (Variables 17 to 24)

FixedIntSubs	InternetUsersPct	GDPPerCapGrowth	GDPGrowth	GDP	PoliticalStability	RuleOfLaw	RqdEduYears
Min. : 0.0000	Min. : 0.00	Min. :-49.12786	Min. :-50.339	Min. :2.355e+07	Min. :-3.31295	Min. :-2.59088	Min. : 4.00
1st Qu.: 0.2673	1st Qu.: 2.81	1st Qu.: 0.07708	1st Qu.: 1.464	1st Qu.:5.577e+09	1st Qu.:-0.69566	1st Qu.:-0.82174	1st Qu.: 8.00
Median : 3.8683	Median : 21.00	Median : 2.23599	Median : 3.755	Median :2.361e+10	Median : 0.02612	Median :-0.20532	Median : 9.00
Mean :10.5358	Mean : 32.58	Mean : 2.19326	Mean : 3.630	Mean :3.372e+11	Mean :-0.06279	Mean :-0.06373	Mean : 9.42
3rd Qu.:18.4017	3rd Qu.: 61.40	3rd Qu.: 4.42206	3rd Qu.: 6.010	3rd Qu.:1.585e+11	3rd Qu.: 0.79477	3rd Qu.: 0.69307	3rd Qu.:11.00
Max. :62.2147	Max. :100.00	Max. :140.49058	Max. :149.973	Max. :2.206e+13	Max. : 1.75868	Max. : 2.12476	Max. :17.00
NA's :2041	NA's :675	NA's :443	NA's :443	NA's :471	NA's :1160	NA's :1114	NA's :1432

Table 4: Summary Statistics (Variables 25 to 32)

GovtEduSpendPctGDP	MortalityFromDirtyness	UHCServiceCoverage	HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct	FoodInsecurityPct	RuralPopulGrowth
Min. : 0.000	Min. : 0.40	Min. :11.00	Min. : 4.176	Min. : 0.1489	Min. : 0.036	Min. : 2.00	Min. :-235.9259
1st Qu.: 3.139	1st Qu.: 2.95	1st Qu.:44.00	1st Qu.: 63.631	1st Qu.: 21.9384	1st Qu.: 1.212	1st Qu.: 8.90	1st Qu.: -0.5789
Median : 4.232	Median : 6.10	Median :63.00	Median : 261.295	Median : 128.2416	Median : 3.663	Median :22.30	Median : 0.4574
Mean : 4.418	Mean : 16.96	Mean :59.39	Mean : 959.978	Mean : 648.7002	Mean : 8.791	Mean :29.09	Mean : 0.3933
3rd Qu.: 5.428	3rd Qu.: 25.55	3rd Qu.:75.00	3rd Qu.: 890.483	3rd Qu.: 557.3557	3rd Qu.:10.632	3rd Qu.:45.30	3rd Qu.: 1.6167
Max. :15.863	Max. :108.10	Max. :91.00	Max. :12434.434	Max. :7871.2450	Max. :88.404	Max. :89.20	Max. : 15.0306
NA's :2288	NA's :5791	NA's :4630	NA's :1617	NA's :1630	NA's :5009	NA's :4953	NA's :458

Table 5: Summary Statistics (Variables 33 to 35)

VoiceAccountability	Homicides	Income Group
Min. :-2.31340	Min. : 0.000	Length:5974
1st Qu.:-0.88313	1st Qu.: 1.228	Class :character
Median :-0.00619	Median : 2.861	Mode :character
Mean :-0.04756	Mean : 7.939	NA
3rd Qu.: 0.85004	3rd Qu.: 8.879	NA
Max. : 1.80099	Max. :138.774	NA
NA's :1112	NA's :2755	NA

Table 6: Data summary

Name	FirstCutData
Number of rows	5974
Number of columns	35
Column type frequency:	
character	3
numeric	32
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
CountryCode	0	1.00	3	9	0	206	0
CountryName	319	0.95	4	38	0	195	0
Income Group	377	0.94	10	19	0	4	0

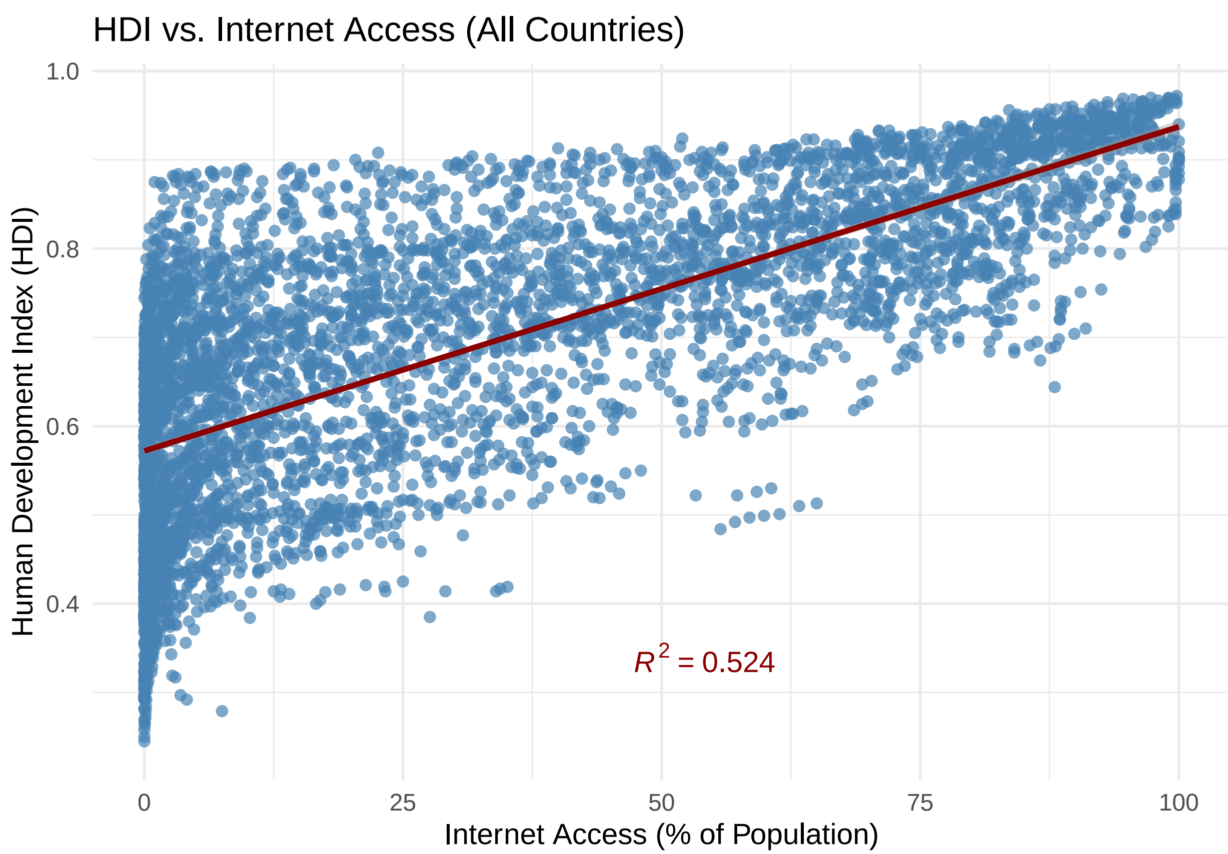
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
year	0	1.00	2.009000e+03	8.370000e+00	1995.00	2002.00	2.009000e+03	2.016000e+03	2.023000e+03	
HDI_Index	422	0.93	6.900000e-01	1.600000e-01	0.24	0.57	7.100000e-01	8.100000e-01	9.700000e-01	
IHDI_Index	3677	0.38	5.800000e-01	2.000000e-01	0.13	0.40	5.900000e-01	7.500000e-01	9.200000e-01	
GiniCoeff	3670	0.39	2.132000e+01	1.135000e+01	4.82	11.06	1.918000e+01	3.188000e+01	5.189000e+01	
FoodIndex	720	0.88	9.133000e+01	2.806000e+01	17.00	76.57	9.452000e+01	1.028000e+02	5.787100e+02	
CorruptionScore	1204	0.80	-6.000000e-02	1.000000e+00	-1.97	-0.82	-3.000000e-01	5.900000e-01	2.460000e+00	
ElectricAccess	627	0.90	7.923000e+01	3.010000e+01	0.80	63.20	9.810000e+01	1.000000e+02	1.000000e+02	
PopDensity	583	0.90	2.982000e+02	1.389800e+03	1.44	30.87	7.792000e+01	1.766100e+02	1.882289e+04	
PopInSlums	4011	0.33	2.873000e+01	2.732000e+01	0.00	1.55	2.118000e+01	5.280000e+01	9.981000e+01	
WaterStress	1505	0.75	6.508000e+01	2.666800e+02	0.03	3.71	1.118000e+01	3.970000e+01	3.850500e+03	
BankingAccess	5412	0.09	5.704000e+01	3.013000e+01	0.40	31.06	5.405000e+01	8.676000e+01	1.000000e+02	
RnDInvest	3666	0.39	9.700000e-01	9.800000e-01	0.01	0.24	6.000000e-01	1.390000e+00	6.020000e+00	
GovtEffectiveness	1223	0.80	-6.000000e-02	1.000000e+00	-2.44	-0.79	-2.000000e-01	5.900000e-01	2.470000e+00	
ShippingIndex	3592	0.40	2.514000e+01	2.522000e+01	0.60	7.54	1.517000e+01	3.416000e+01	1.711800e+02	
FixedIntSubs	2041	0.66	1.054000e+01	1.307000e+01	0.00	0.27	3.870000e+00	1.840000e+01	6.221000e+01	
InternetUsersPct	675	0.89	3.258000e+01	3.211000e+01	0.00	2.81	2.100000e+01	6.140000e+01	1.000000e+02	
GDPPerCapGrowth	443	0.93	2.190000e+00	6.020000e+00	-49.13	0.08	2.240000e+00	4.420000e+00	1.404900e+02	
GDPGrowth	443	0.93	3.630000e+00	6.160000e+00	-50.34	1.46	3.760000e+00	6.010000e+00	1.499700e+02	
GDP	471	0.92	3.371605e+11	1.471154e+12	23553249.10	5577146682.40	2.360523e+10	1.584701e+11	2.206258e+13	
PoliticalStability	1160	0.81	-6.000000e-02	1.000000e+00	-3.31	-0.70	3.000000e-02	7.900000e-01	1.760000e+00	
RuleOfLaw	1114	0.81	-6.000000e-02	9.900000e-01	-2.59	-0.82	-2.100000e-01	6.900000e-01	2.120000e+00	
RqdEduYears	1432	0.76	9.420000e+00	2.250000e+00	4.00	8.00	9.000000e+00	1.100000e+01	1.700000e+01	
GovtEduSpendPctGDP	2288	0.62	4.420000e+00	1.890000e+00	0.00	3.14	4.230000e+00	5.430000e+00	1.586000e+01	
MortalityFromDirtyness	5791	0.03	1.696000e+01	2.165000e+01	0.40	2.95	6.100000e+00	2.555000e+01	1.081000e+02	
UHCServiceCoverage	4630	0.22	5.939000e+01	1.902000e+01	11.00	44.00	6.300000e+01	7.500000e+01	9.100000e+01	
HealthSpendPerCapita	1617	0.73	9.599800e+02	1.689180e+03	4.18	63.63	2.612900e+02	8.904800e+02	1.243443e+04	
GovtHealthSpendPerCapita	1630	0.73	6.487000e+02	1.235080e+03	0.15	21.94	1.282400e+02	5.573600e+02	7.871240e+03	
MigrantsPct	5009	0.16	8.790000e+00	1.358000e+01	0.04	1.21	3.660000e+00	1.063000e+01	8.840000e+01	
FoodInsecurityPct	4953	0.17	2.909000e+01	2.301000e+01	2.00	8.90	2.230000e+01	4.530000e+01	8.920000e+01	

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
RuralPopulGrowth	458	0.92	3.900000e-01	3.700000e+00	-235.93	-0.58	4.600000e-01	1.620000e+00	1.503000e+01	
VoiceAccountability	1112	0.81	-5.000000e-02	1.000000e+00	-2.31	-0.88	-1.000000e-02	8.500000e-01	1.800000e+00	
Homicides	2755	0.54	7.940000e+00	1.247000e+01	0.00	1.23	2.860000e+00	8.880000e+00	1.387700e+02	

```r
# Filter complete cases again (just in case)
ggplot(FirstCutData, aes(x = InternetUsersPct, y = HDI_Index)) +
  geom_point(alpha = 0.7, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "darkred") +
  stat_poly_eq(
    aes(label = after_stat(rr.label)),
    formula = y ~ x,
    parse = TRUE,
    output.type = "expression",
    rr.digits = 3,
    label.x = 0.6,           # X-position (adjust if needed)
    label.y = 0.15,          # Y-position (bottom-ish)
    color = "darkred",       # Match regression line
    na.rm = TRUE
  ) +
  labs(
    title = "HDI vs. Internet Access (All Countries)",
    x = "Internet Access (% of Population)",
    y = "Human Development Index (HDI)"
  ) +
  theme_minimal()
```

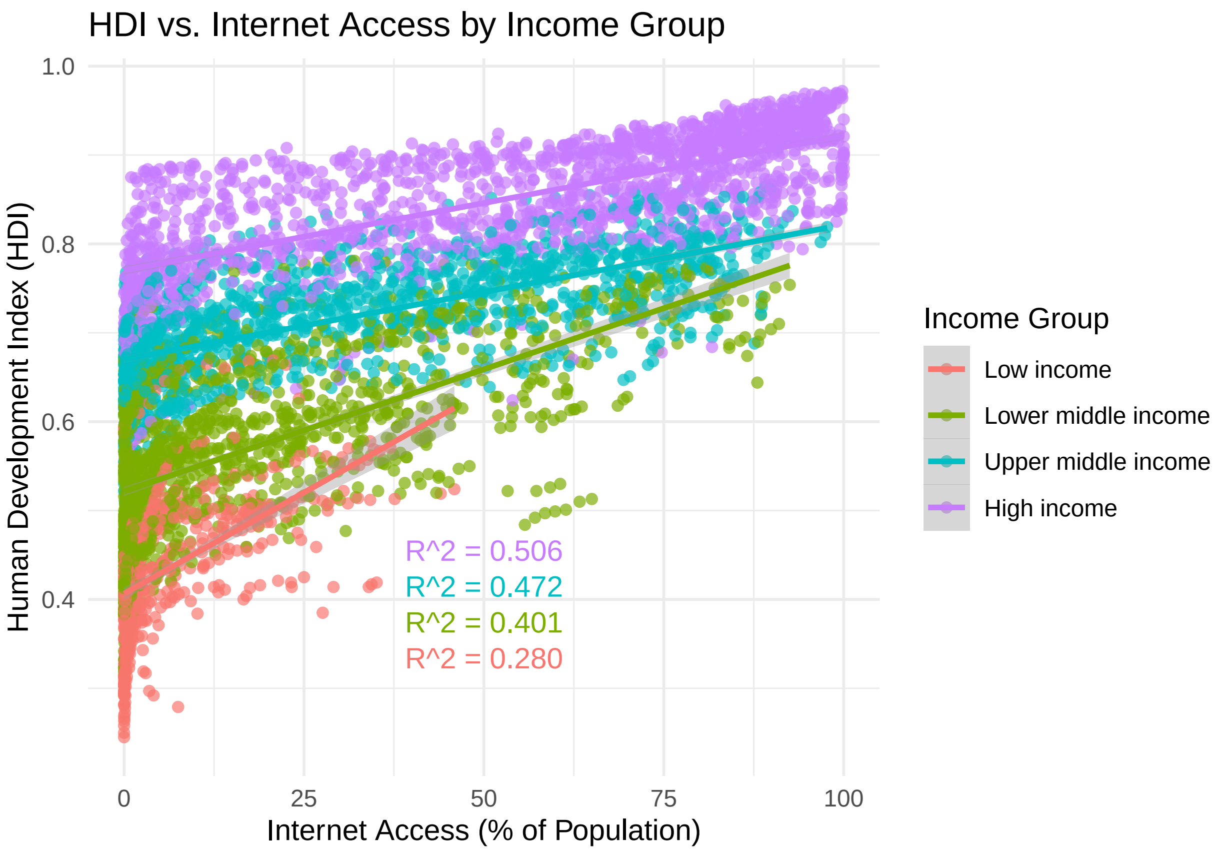
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 978 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 978 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
FirstCutDataFiltered <- FirstCutData[!is.na(FirstCutData$`Income Group`), ]
FirstCutDataFiltered$`Income Group` <- factor(
  FirstCutDataFiltered$`Income Group`,
  levels = c("Low income", "Lower middle income", "Upper middle income", "High income")
)
ggplot(FirstCutDataFiltered, aes(x = InternetUsersPct, y = HDI_Index, color = `Income Group`)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE) +
  stat_poly_eq(
    aes(
      label = after_stat(rr.label),
      group = `Income Group`,
```

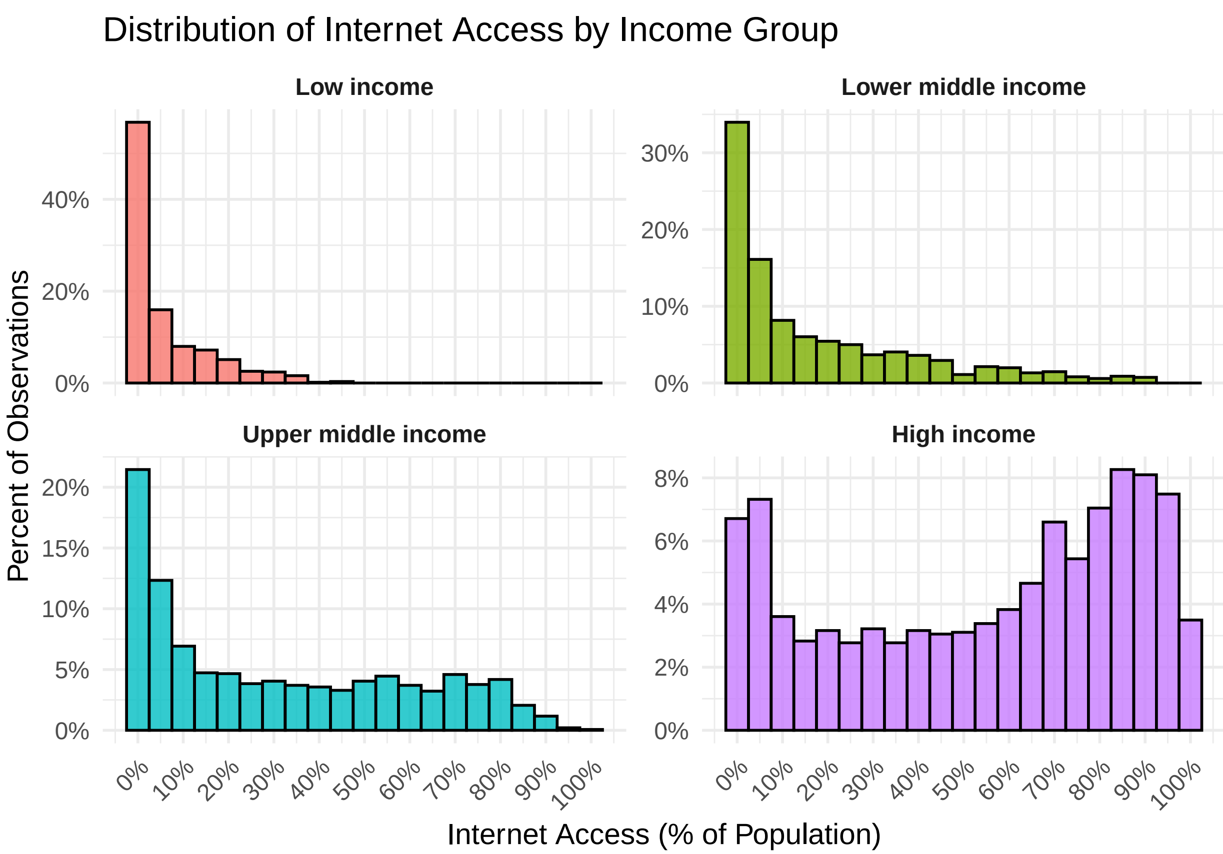
```r
      color = `Income Group`
    ),
    formula = y ~ x,
    parse = FALSE,
    output.type = "text",
    rr.digits = 3,
    label.x = 0.6,
    label.y = c(0.15, 0.2, 0.25, 0.3),  # Stack in bottom right
    na.rm = TRUE
  ) +
  labs(
    title = "HDI vs. Internet Access by Income Group",
    x = "Internet Access (% of Population)",
    y = "Human Development Index (HDI)",
    color = "Income Group"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 646 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 646 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



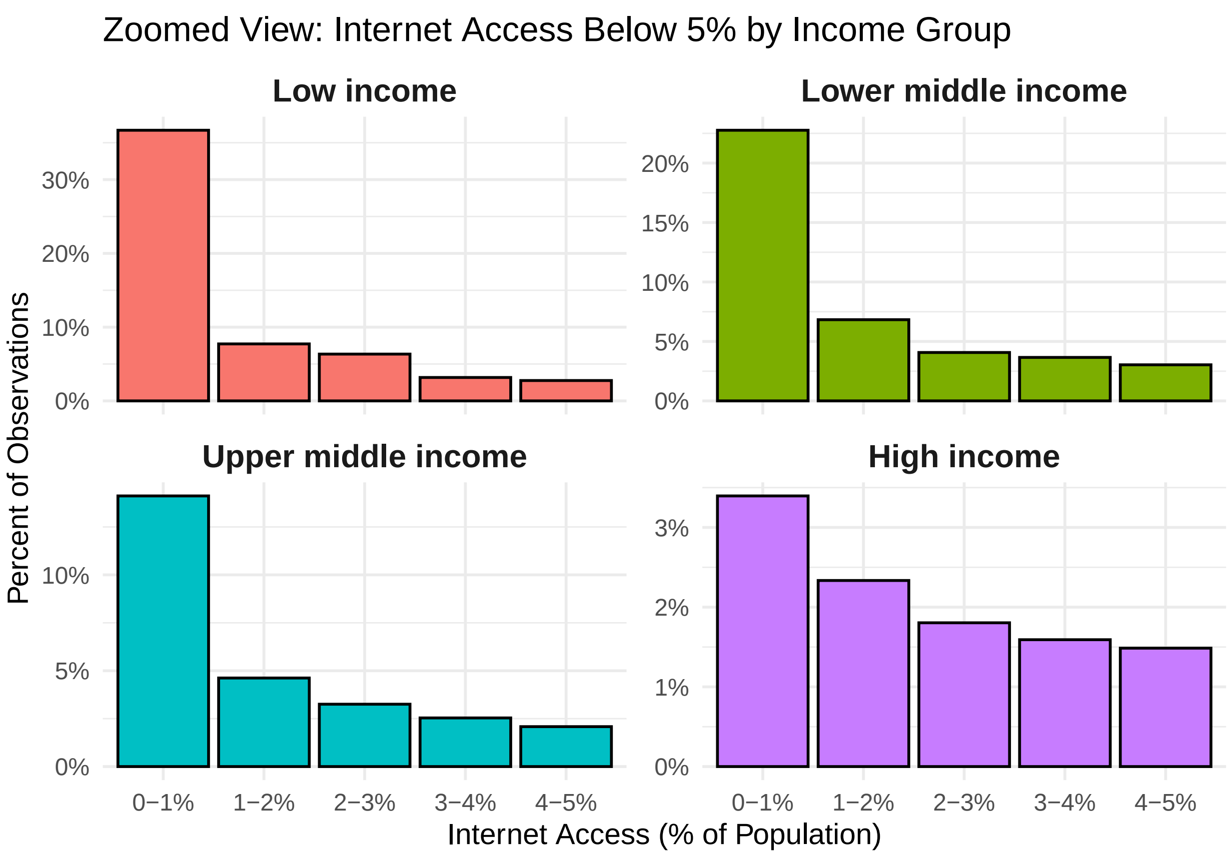

```r
FirstCutDataFiltered <- FirstCutData[!is.na(FirstCutData$`Income Group`), ]

FirstCutDataFiltered$`Income Group` <- factor(
  FirstCutDataFiltered$`Income Group`,
  levels = c("Low income", "Lower middle income", "Upper middle income", "High income")
)

FirstCutDataFilteredClean <- FirstCutDataFiltered %>%
  filter(!is.na(InternetUsersPct))


ggplot(FirstCutDataFilteredClean, aes(x = InternetUsersPct, fill = `Income Group`)) +
```

```r
geom_histogram(
    aes(y = after_stat(count / tapply(count, PANEL, sum)[as.integer(PANEL)])),
    binwidth = 5,
    color = "black",
    alpha = 0.8
  ) +
  facet_wrap(~ `Income Group`, scales = "free_y") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  scale_x_continuous(
    breaks = seq(0, 100, by = 10),
    labels = function(x) paste0(x, "%")
  ) +
  labs(
    title = "Distribution of Internet Access by Income Group",
    x = "Internet Access (% of Population)",
    y = "Percent of Observations"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold"),
    axis.text.x = element_text(angle = 45, hjust = 1),
    legend.position = "none"
  )
```




```
# Step 1: Filter to only rows with Internet Access under 5%
# Bin and summarize within the 0%-5% range
# Bin and summarize within the 0%-5% range
zoom_data <- FirstCutDataFiltered %>%
  filter(InternetUsersPct >= 0, InternetUsersPct < 5) %>%
  mutate(Bin = cut(
    InternetUsersPct,
    breaks = seq(0, 5, by = 1),
    include.lowest = TRUE,
    right = FALSE,
    labels = c("0-1%", "1-2%", "2-3%", "3-4%", "4-5%")
  ))
```

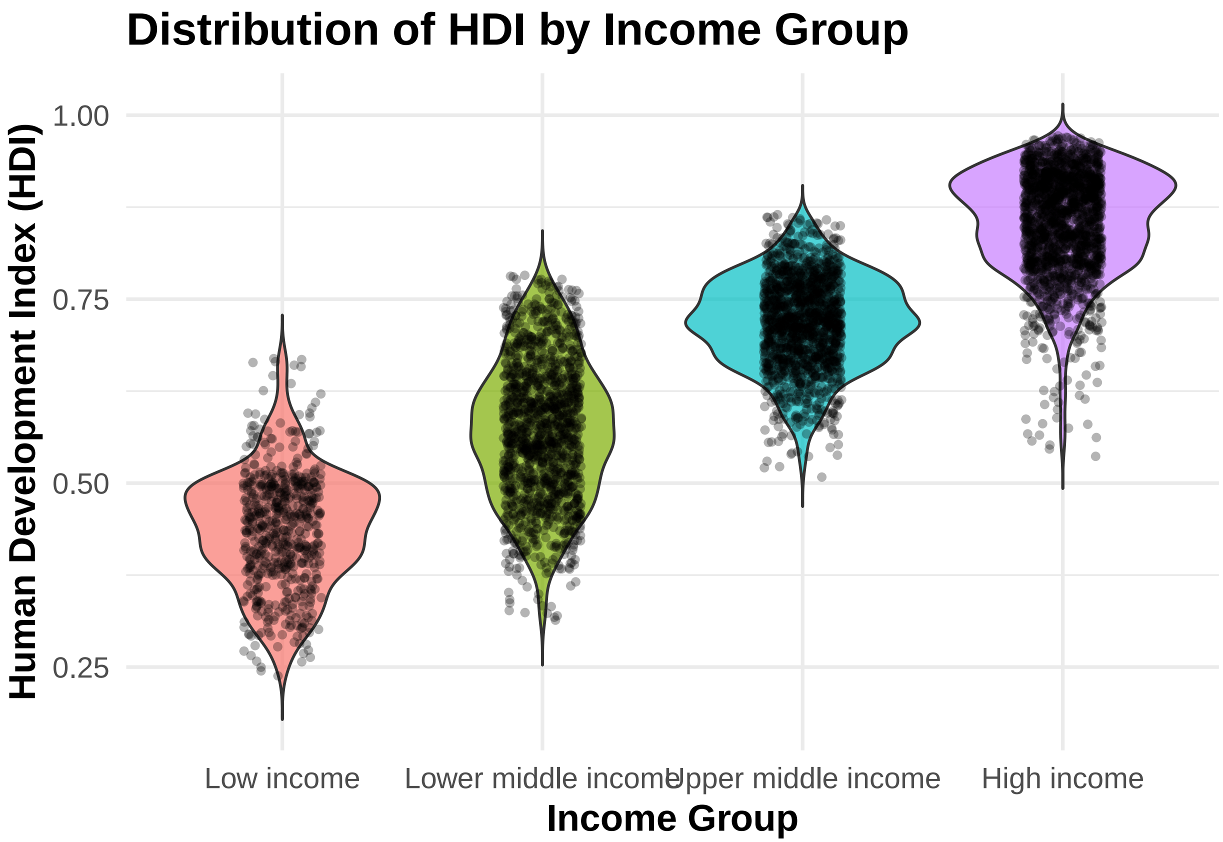
```r
# Group totals for each Income Group (full data)
group_totals <- FirstCutDataFiltered %>%
  count(`Income Group`, name = "GroupTotal")

# Bin counts and percentages
zoom_summary <- zoom_data %>%
  count(`Income Group`, Bin, name = "BinCount") %>%
  left_join(group_totals, by = "Income Group") %>%
  mutate(Percent = BinCount / GroupTotal)

# Plot
ggplot(zoom_summary, aes(x = Bin, y = Percent, fill = `Income Group`)) +
  geom_col(color = "black", width = 0.9) +
  facet_wrap(~ `Income Group`, scales = "free_y") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(
    title = "Zoomed View: Internet Access Below 5% by Income Group",
    x = "Internet Access (% of Population)",
    y = "Percent of Observations"
  ) +
  theme_minimal() +
  theme(
    strip.text = element_text(face = "bold", size = 12),
    axis.text.x = element_text(angle = 0, hjust = 0.5),
    legend.position = "none"
  )
```




```r
ggplot(FirstCutDataFiltered, aes(x = `Income Group`, y = HDI_Index, fill = `Income Group`)) +
  geom_violin(trim = FALSE, alpha = 0.7) +
  geom_jitter(width = 0.15, alpha = 0.3, color = "black", size = 1) +
  labs(
    title = "Distribution of HDI by Income Group",
    x = "Income Group",
    y = "Human Development Index (HDI)"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "none",
    plot.title = element_text(face = "bold"),
```

```
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold")
  )
```

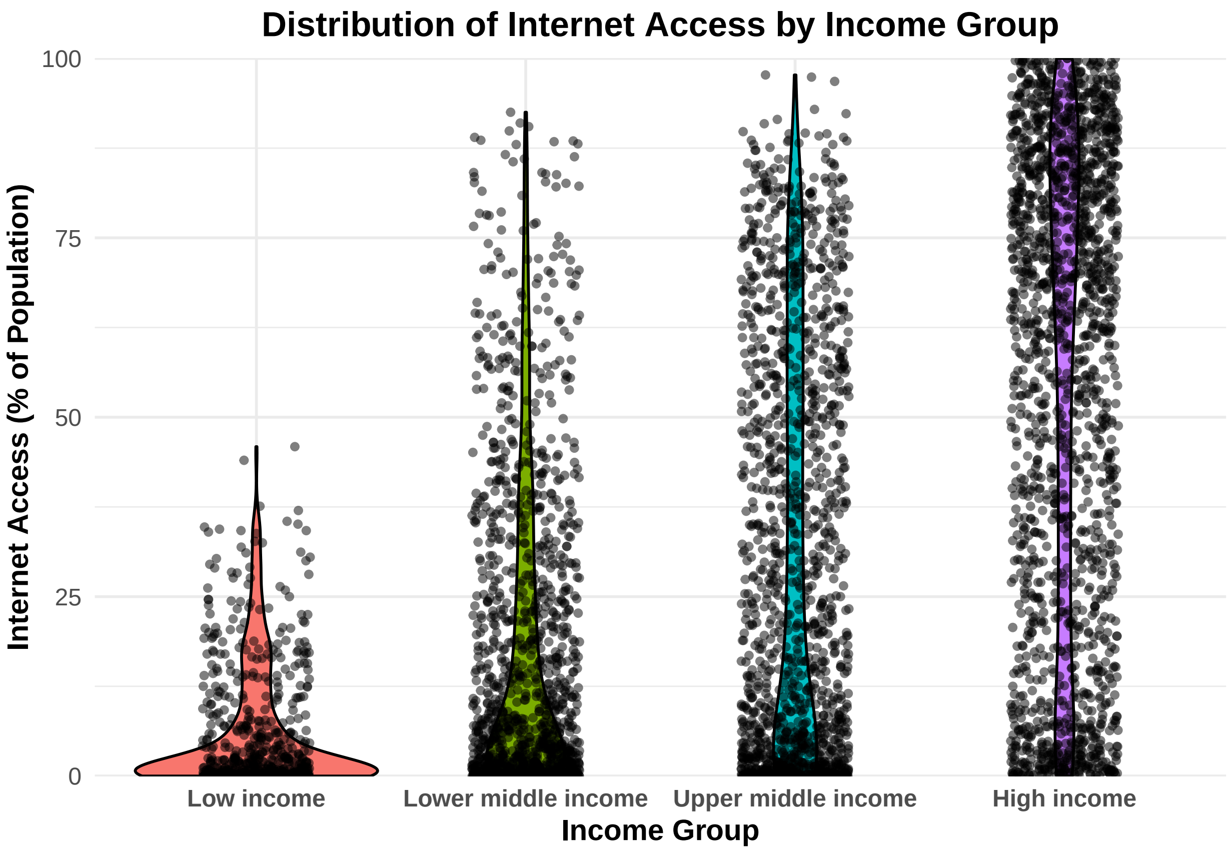
```
## Warning: Removed 417 rows containing non-finite outside the scale range
## (`stat_ydensity()`).

## Warning: Removed 417 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggplot(FirstCutDataFiltered, aes(x = `Income Group`, y = InternetUsersPct, fill = `Income Group`)) +
  geom_violin(trim = TRUE, color = "black", adjust = 1.2) +  # trim removes tails beyond the data
  geom_jitter(width = 0.2, size = 1, alpha = 0.5, color = "black") +
```

```r
  scale_y_continuous(limits = c(0, 100), expand = c(0, 0)) +  # Hard clip y-axis
  labs(
    title = "Distribution of Internet Access by Income Group",
    y = "Internet Access (% of Population)",
    x = "Income Group"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold"),
    axis.title = element_text(face = "bold"),
    axis.text.x = element_text(face = "bold"),
    legend.position = "none"
  ) +
  scale_fill_manual(values = c(
    "Low income" = "#F8766D",
    "Lower middle income" = "#7CAE00",
    "Upper middle income" = "#00BFC4",
    "High income" = "#C77CFF"
  ))
```

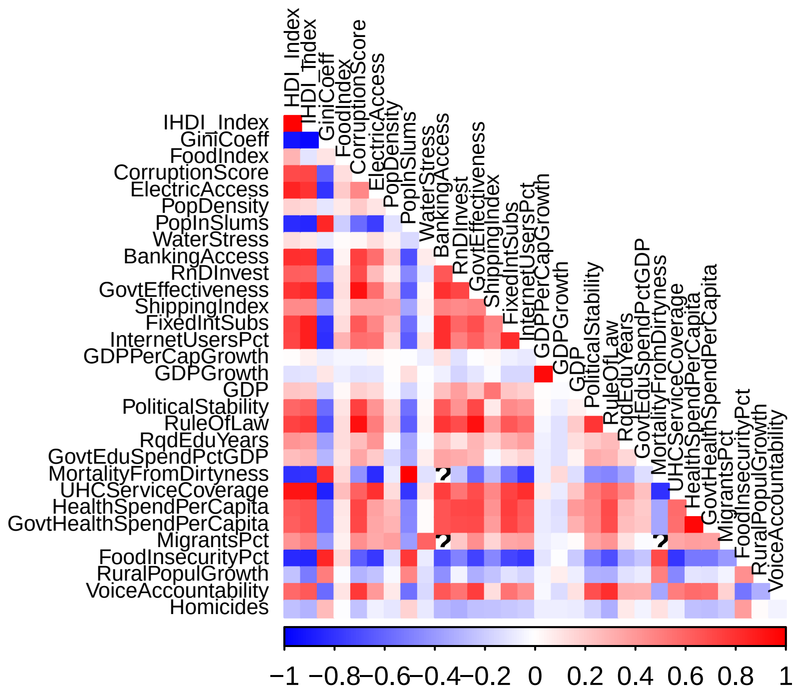
```
## Warning: Removed 348 rows containing non-finite outside the scale range
## (`stat_ydensity()`).

## Warning: Removed 368 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```r
# Step 1: Select only numeric columns and drop unwanted ones
numeric_vars <- FirstCutData %>%
  select(where(is.numeric)) %>%
  select(-year)  # Drop the 'Year' column

# Step 2: Compute correlation matrix using pairwise complete observations
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs", method = "pearson")
cor_matrix_90 <- ifelse(abs(cor_matrix) >= 0.9, cor_matrix, NA)
cor_matrix_70_90 <- ifelse(abs(cor_matrix) >= 0.7 & abs(cor_matrix) < 0.9, cor_matrix, NA)


# Step 3: Plot correlation heatmap
```

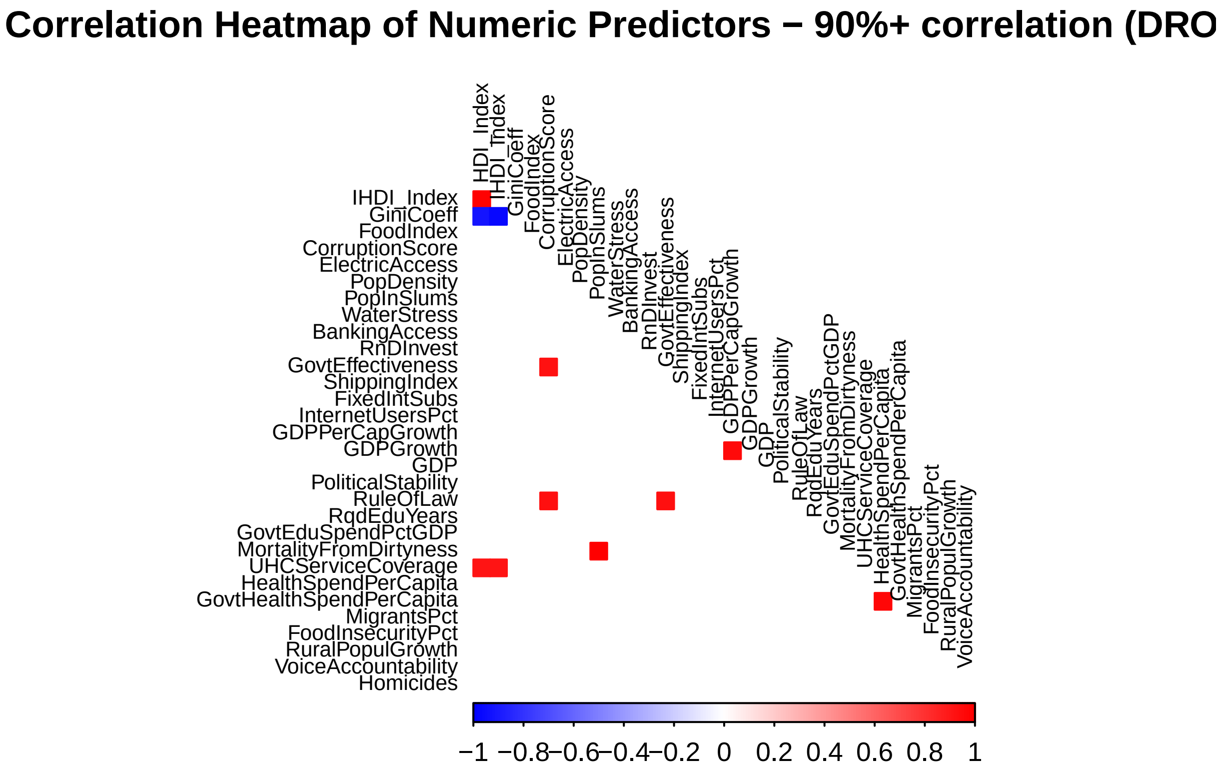
```r
corrplot(cor_matrix,
         method = "color",
         type = "lower",
         tl.col = "black",
         tl.cex = 0.7,
         col = colorRampPalette(c("blue", "white", "red"))(200),
         diag = FALSE)
title("Correlation Heatmap of Numeric Predictors - ALL", line = 1, cex.main = 1.2)
```

Correlation Heatmap of Numeric Predictors – ALL



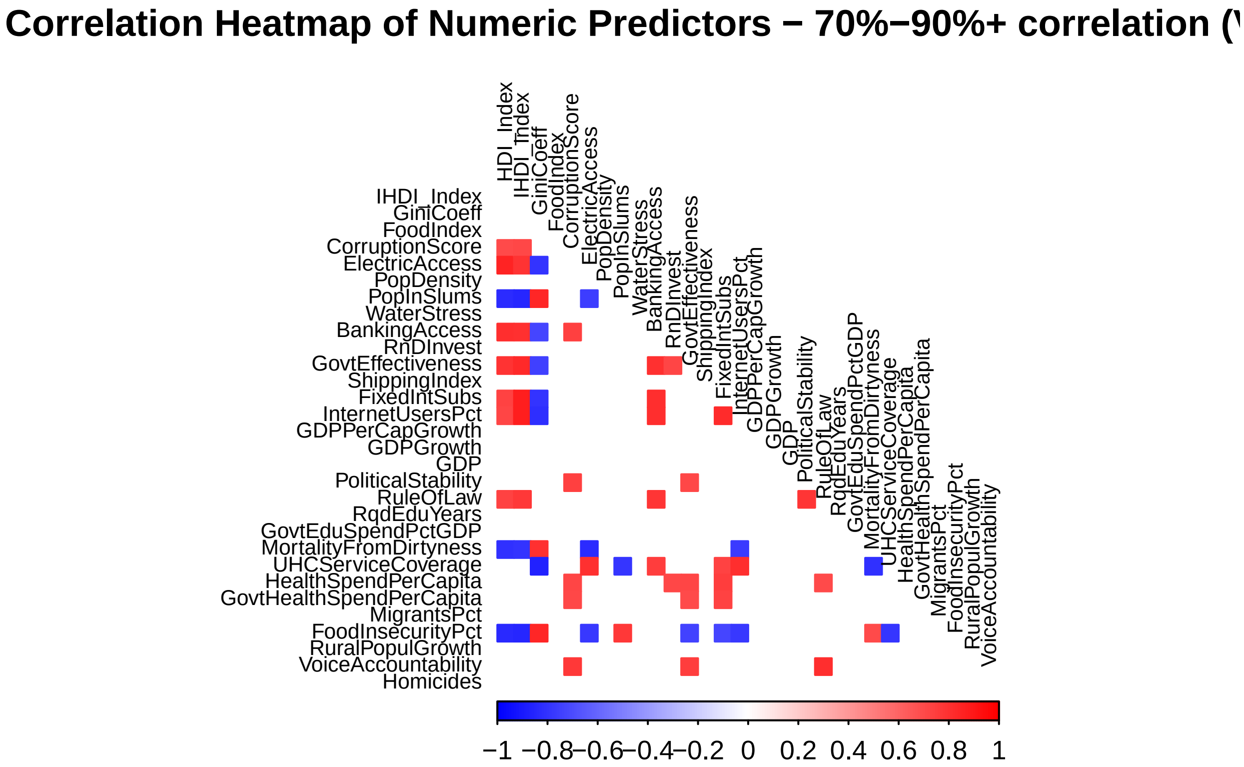

```r
# Plot only strong correlations
corrplot(cor_matrix_90,
         method = "color",
         type = "lower",
         tl.col = "black",
```

```r
        tl.cex = 0.7,
        col = colorRampPalette(c("blue", "white", "red"))(200),
        diag = FALSE,
        na.label = " ")  # blank for missing (non-displayed) values
title("Correlation Heatmap of Numeric Predictors - 90%+ correlation (DROP)", line = 1, cex.main = 1.2)
```




```r
corrplot(cor_matrix_70_90,
        method = "color",
        type = "lower",
        tl.col = "black",
        tl.cex = 0.7,
        col = colorRampPalette(c("blue", "white", "red"))(200),
        diag = FALSE,
        na.label = " ")  # blank for missing (non-displayed) values
```

```r
title("Correlation Heatmap of Numeric Predictors - 70%-90%+ correlation (VIF)", line = 1, cex.main = 1.2)
```




```r
# Step 2: Compute Pearson correlation matrix using pairwise complete observations
cor_matrix <- cor(numeric_vars, use = "pairwise.complete.obs", method = "pearson")

# Step 3 Print
# Get total number of columns
cols <- ncol(cor_matrix)
col_names <- colnames(cor_matrix)

# Loop through and print in chunks
for (i in seq(1, cols, by = cols_per_table)) {
  # Subset the matrix
  end_col <- min(i + cols_per_table - 1, cols)
```

```r
  sub_matrix <- cor_matrix[, i:end_col]

  # Print each chunk with a dynamic caption
  cat(
    asis_output(
      kable(
        round(sub_matrix, 2),
        format = "latex",
        booktabs = TRUE,
        caption = paste0("Correlation Matrix: Columns ", i, " to ", end_col)) %>%
        kable_styling(latex_options = c("HOLD_Position", "striped", "scale_down"))
      )
  )
}
```

```r
# Step 1: Compute % missing per variable per year
missing_heatmap_data <- FirstCutData %>%
  group_by(year) %>%
  summarise(across(everything(), ~mean(is.na(.)) * 100)) %>%
  pivot_longer(-year, names_to = "variable", values_to = "pct_missing")

# Step 2: Plot with stoplight colors
stoplight <- c("#1a9641", "#ffea00", "#d7191c")
print(ggplot(missing_heatmap_data, aes(x = year, y = variable, fill = pct_missing)) +
  geom_tile(color = "white") +
  scale_fill_gradientn(
    colours = stoplight,
    values = scales::rescale(c(0, 25, 100)),
    limits = c(0, 100),
    name = "% Missing"
  ) +
  labs(
    title = "Missing Data Heatmap",
    x = "Year",
    y = "Variable"
  ) +
  theme_minimal(base_size = 11) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
```

Table 9: Correlation Matrix: Columns 1 to 8

	HDI_Index	IHDI_Index	GiniCoeff	FoodIndex	CorruptionScore	ElectricAccess	PopDensity	PopInSlums
HDI_Index	1.00	0.98	-0.92	0.29	0.71	0.85	0.17	-0.82
IHDI_Index	0.98	1.00	-0.97	-0.10	0.72	0.80	0.14	-0.85
GiniCoeff	-0.92	-0.97	1.00	0.10	-0.62	-0.79	-0.09	0.84
FoodIndex	0.29	-0.10	0.10	1.00	0.13	0.20	0.08	-0.20
CorruptionScore	0.71	0.72	-0.62	0.13	1.00	0.47	0.20	-0.57
ElectricAccess	0.85	0.80	-0.79	0.20	0.47	1.00	0.10	-0.75
PopDensity	0.17	0.14	-0.09	0.08	0.20	0.10	1.00	-0.11
PopInSlums	-0.82	-0.85	0.84	-0.20	-0.57	-0.75	-0.11	1.00
WaterStress	0.12	0.08	-0.08	0.01	0.04	0.13	0.05	-0.14
BankingAccess	0.82	0.81	-0.72	0.04	0.75	0.56	0.18	-0.69
RnDInvest	0.63	0.62	-0.47	0.12	0.69	0.27	0.07	-0.47
GovtEffectiveness	0.79	0.83	-0.74	0.12	0.92	0.57	0.22	-0.64
ShippingIndex	0.46	0.46	-0.38	0.10	0.35	0.34	0.34	-0.36
FixedIntSubs	0.73	0.88	-0.80	0.14	0.64	0.47	0.24	-0.57
InternetUsersPct	0.72	0.87	-0.82	0.29	0.56	0.55	0.15	-0.64
GDPPerCapGrowth	0.00	0.05	-0.06	-0.03	-0.03	0.03	0.01	0.00
GDPGrowth	-0.12	-0.10	0.10	-0.07	-0.10	-0.09	0.00	0.12
GDP	0.22	0.21	-0.17	0.03	0.19	0.14	-0.02	-0.17
PoliticalStability	0.59	0.63	-0.58	0.10	0.75	0.41	0.14	-0.56
RuleOfLaw	0.74	0.77	-0.68	0.13	0.94	0.50	0.18	-0.63
RqdEduYears	0.41	0.39	-0.38	0.13	0.26	0.42	-0.01	-0.36
GovtEduSpendPctGDP	0.26	0.28	-0.28	0.11	0.35	0.20	-0.15	-0.33
MortalityFromDirtyness	-0.80	-0.79	0.80	0.15	-0.42	-0.82	-0.07	1.00
UHCServiceCoverage	0.92	0.92	-0.86	0.25	0.61	0.81	0.13	-0.79
HealthSpendPerCapita	0.64	0.68	-0.57	0.09	0.73	0.36	0.26	-0.48
GovtHealthSpendPerCapita	0.63	0.67	-0.57	0.09	0.72	0.35	0.29	-0.47
MigrantsPct	0.41	0.50	-0.41	0.06	0.43	0.34	0.37	-0.38
FoodInsecurityPct	-0.83	-0.85	0.84	0.14	-0.62	-0.78	-0.13	0.77
RuralPopulGrowth	-0.24	-0.52	0.50	-0.02	-0.30	-0.23	-0.03	0.48
VoiceAccountability	0.59	0.65	-0.58	0.10	0.77	0.40	0.08	-0.55
Homicides	-0.24	-0.30	0.26	-0.01	-0.24	-0.06	-0.09	0.17

Table 10: Correlation Matrix: Columns 9 to 16

	WaterStress	BankingAccess	RnDInvest	GovtEffectiveness	ShippingIndex	FixedIntSubs	InternetUsersPct	GDPPerCapGrowth
HDI_Index	0.12	0.82	0.63	0.79	0.46	0.73	0.72	0.00
IHDI_Index	0.08	0.81	0.62	0.83	0.46	0.88	0.87	0.05
GiniCoeff	-0.08	-0.72	-0.47	-0.74	-0.38	-0.80	-0.82	-0.06
FoodIndex	0.01	0.04	0.12	0.12	0.10	0.14	0.29	-0.03
CorruptionScore	0.04	0.75	0.69	0.92	0.35	0.64	0.56	-0.03
ElectricAccess	0.13	0.56	0.27	0.57	0.34	0.47	0.55	0.03
PopDensity	0.05	0.18	0.07	0.22	0.34	0.24	0.15	0.01
PopInSlums	-0.14	-0.69	-0.47	-0.64	-0.36	-0.57	-0.64	0.00
WaterStress	1.00	0.08	-0.08	0.04	0.05	-0.04	0.10	-0.07
BankingAccess	0.08	1.00	0.66	0.80	0.49	0.81	0.81	0.12
RnDInvest	-0.08	0.66	1.00	0.72	0.46	0.59	0.49	-0.12
GovtEffectiveness	0.04	0.80	0.72	1.00	0.49	0.69	0.62	-0.01
ShippingIndex	0.05	0.49	0.46	0.49	1.00	0.49	0.46	0.03
FixedIntSubs	-0.04	0.81	0.59	0.69	0.49	1.00	0.83	-0.06
InternetUsersPct	0.10	0.81	0.49	0.62	0.46	0.83	1.00	-0.08
GDPPerCapGrowth	-0.07	0.12	-0.12	-0.01	0.03	-0.06	-0.08	1.00
GDPGrowth	-0.01	-0.05	-0.16	-0.09	-0.01	-0.16	-0.16	0.96
GDP	-0.01	0.24	0.37	0.23	0.53	0.22	0.19	0.01
PoliticalStability	0.03	0.65	0.43	0.71	0.09	0.46	0.41	0.01
RuleOfLaw	0.05	0.78	0.70	0.94	0.39	0.65	0.57	-0.03
RqdEduYears	-0.02	0.23	0.11	0.28	0.16	0.32	0.38	-0.05
GovtEduSpendPctGDP	0.06	0.36	0.32	0.27	-0.02	0.12	0.17	-0.07
MortalityFromDirtyness	-0.12	NA	-0.23	-0.57	-0.27	-0.56	-0.76	-0.06
UHCServiceCoverage	0.10	0.76	0.54	0.70	0.46	0.73	0.82	0.07
HealthSpendPerCapita	0.01	0.66	0.71	0.72	0.43	0.75	0.64	-0.07
GovtHealthSpendPerCapita	0.02	0.66	0.69	0.71	0.39	0.74	0.63	-0.07
MigrantsPct	0.61	NA	0.21	0.43	0.17	0.35	0.37	-0.06
FoodInsecurityPct	-0.08	-0.69	-0.49	-0.74	-0.47	-0.72	-0.77	-0.11
RuralPopulGrowth	-0.13	-0.44	-0.05	-0.31	-0.24	-0.12	-0.16	-0.03
VoiceAccountability	-0.14	0.65	0.54	0.75	0.13	0.54	0.44	-0.02
Homicides	-0.09	-0.28	-0.31	-0.24	-0.23	-0.22	-0.20	-0.06

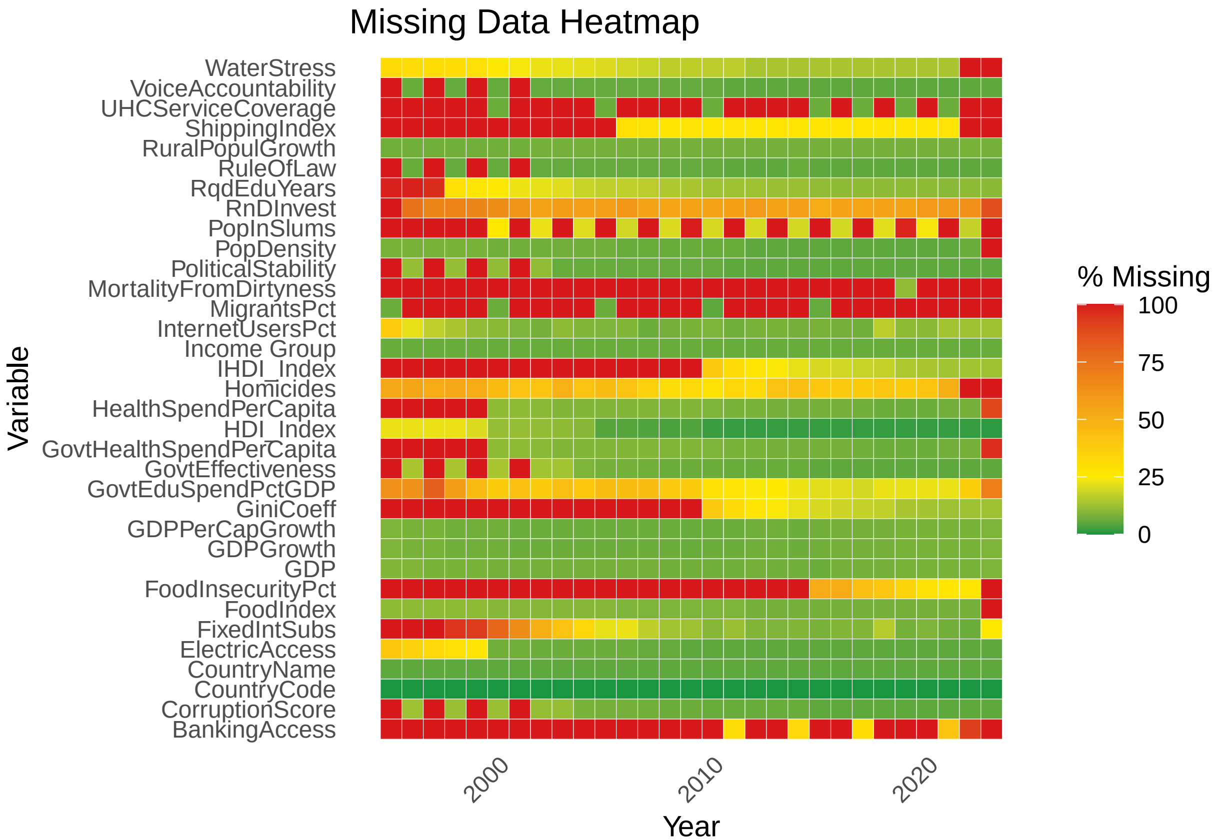
Table 11: Correlation Matrix: Columns 17 to 24

	GDPGrowth	GDP	PoliticalStability	RuleOfLaw	RqdEduYears	GovtEduSpendPctGDP	MortalityFromDirtyness	UHCServiceCoverage
HDI_Index	-0.12	0.22	0.59	0.74	0.41	0.26	-0.80	0.92
IHDI_Index	-0.10	0.21	0.63	0.77	0.39	0.28	-0.79	0.92
GiniCoeff	0.10	-0.17	-0.58	-0.68	-0.38	-0.28	0.80	-0.86
FoodIndex	-0.07	0.03	0.10	0.13	0.13	0.11	0.15	0.25
CorruptionScore	-0.10	0.19	0.75	0.94	0.26	0.35	-0.42	0.61
ElectricAccess	-0.09	0.14	0.41	0.50	0.42	0.20	-0.82	0.81
PopDensity	0.00	-0.02	0.14	0.18	-0.01	-0.15	-0.07	0.13
PopInSlums	0.12	-0.17	-0.56	-0.63	-0.36	-0.33	1.00	-0.79
WaterStress	-0.01	-0.01	0.03	0.05	-0.02	0.06	-0.12	0.10
BankingAccess	-0.05	0.24	0.65	0.78	0.23	0.36	NA	0.76
RnDInvest	-0.16	0.37	0.43	0.70	0.11	0.32	-0.23	0.54
GovtEffectiveness	-0.09	0.23	0.71	0.94	0.28	0.27	-0.57	0.70
ShippingIndex	-0.01	0.53	0.09	0.39	0.16	-0.02	-0.27	0.46
FixedIntSubs	-0.16	0.22	0.46	0.65	0.32	0.12	-0.56	0.73
InternetUsersPct	-0.16	0.19	0.41	0.57	0.38	0.17	-0.76	0.82
GDPPerCapGrowth	0.96	0.01	0.01	-0.03	-0.05	-0.07	-0.06	0.07
GDPGrowth	1.00	-0.02	-0.06	-0.12	-0.11	-0.11	0.14	-0.08
GDP	-0.02	1.00	0.05	0.20	0.13	0.05	-0.12	0.22
PoliticalStability	-0.06	0.05	1.00	0.78	0.20	0.33	-0.45	0.51
RuleOfLaw	-0.12	0.20	0.78	1.00	0.26	0.29	-0.47	0.62
RqdEduYears	-0.11	0.13	0.20	0.26	1.00	0.16	-0.35	0.45
GovtEduSpendPctGDP	-0.11	0.05	0.33	0.29	0.16	1.00	-0.13	0.25
MortalityFromDirtyness	0.14	-0.12	-0.45	-0.47	-0.35	-0.13	1.00	-0.80
UHCServiceCoverage	-0.08	0.22	0.51	0.62	0.45	0.25	-0.80	1.00
HealthSpendPerCapita	-0.12	0.40	0.46	0.71	0.28	0.20	-0.35	0.58
GovtHealthSpendPerCapita	-0.12	0.31	0.46	0.70	0.26	0.21	-0.34	0.56
MigrantsPct	-0.03	0.01	0.36	0.42	0.11	-0.01	NA	0.35
FoodInsecurityPct	0.00	-0.20	-0.51	-0.68	-0.30	-0.22	0.70	-0.79
RuralPopulGrowth	0.07	-0.07	-0.31	-0.31	-0.12	-0.08	0.49	-0.46
VoiceAccountability	-0.13	0.11	0.68	0.82	0.32	0.31	-0.34	0.50
Homicides	-0.07	-0.08	-0.18	-0.31	0.09	-0.04	0.11	-0.07

Table 12: Correlation Matrix: Columns 25 to 31

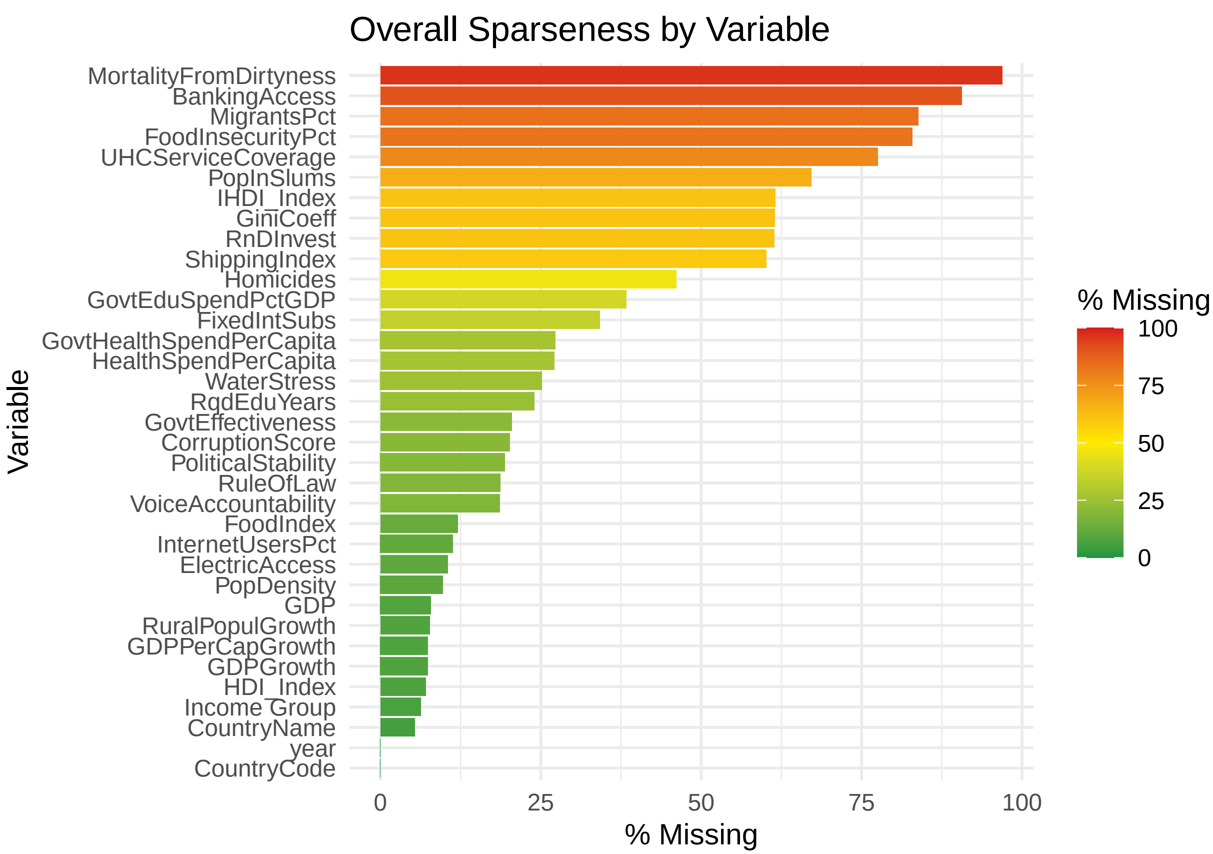
	HealthSpendPerCapita	GovtHealthSpendPerCapita	MigrantsPct	FoodInsecurityPct	RuralPopulGrowth	VoiceAccountability	Homicides
HDI_Index	0.64	0.63	0.41	-0.83	-0.24	0.59	-0.24
IHDI_Index	0.68	0.67	0.50	-0.85	-0.52	0.65	-0.30
GiniCoeff	-0.57	-0.57	-0.41	0.84	0.50	-0.58	0.26
FoodIndex	0.09	0.09	0.06	0.14	-0.02	0.10	-0.01
CorruptionScore	0.73	0.72	0.43	-0.62	-0.30	0.77	-0.24
ElectricAccess	0.36	0.35	0.34	-0.78	-0.23	0.40	-0.06
PopDensity	0.26	0.29	0.37	-0.13	-0.03	0.08	-0.09
PopInSlums	-0.48	-0.47	-0.38	0.77	0.48	-0.55	0.17
WaterStress	0.01	0.02	0.61	-0.08	-0.13	-0.14	-0.09
BankingAccess	0.66	0.66	NA	-0.69	-0.44	0.65	-0.28
RnDInvest	0.71	0.69	0.21	-0.49	-0.05	0.54	-0.31
GovtEffectiveness	0.72	0.71	0.43	-0.74	-0.31	0.75	-0.24
ShippingIndex	0.43	0.39	0.17	-0.47	-0.24	0.13	-0.23
FixedIntSubs	0.75	0.74	0.35	-0.72	-0.12	0.54	-0.22
InternetUsersPct	0.64	0.63	0.37	-0.77	-0.16	0.44	-0.20
GDPPerCapGrowth	-0.07	-0.07	-0.06	-0.11	-0.03	-0.02	-0.06
GDPGrowth	-0.12	-0.12	-0.03	0.00	0.07	-0.13	-0.07
GDP	0.40	0.31	0.01	-0.20	-0.07	0.11	-0.08
PoliticalStability	0.46	0.46	0.36	-0.51	-0.31	0.68	-0.18
RuleOfLaw	0.71	0.70	0.42	-0.68	-0.31	0.82	-0.31
RqdEduYears	0.28	0.26	0.11	-0.30	-0.12	0.32	0.09
GovtEduSpendPctGDP	0.20	0.21	-0.01	-0.22	-0.08	0.31	-0.04
MortalityFromDirtyness	-0.35	-0.34	NA	0.70	0.49	-0.34	0.11
UHCServiceCoverage	0.58	0.56	0.35	-0.79	-0.46	0.50	-0.07
HealthSpendPerCapita	1.00	0.97	0.38	-0.52	-0.10	0.57	-0.24
GovtHealthSpendPerCapita	0.97	1.00	0.37	-0.53	-0.11	0.56	-0.26
MigrantsPct	0.38	0.37	1.00	-0.39	-0.05	0.17	-0.21
FoodInsecurityPct	-0.52	-0.53	-0.39	1.00	0.43	-0.54	0.39
RuralPopulGrowth	-0.10	-0.11	-0.05	0.43	1.00	-0.31	0.01
VoiceAccountability	0.57	0.56	0.17	-0.54	-0.31	1.00	-0.04
Homicides	-0.24	-0.26	-0.21	0.39	0.01	-0.04	1.00

```
    panel.grid = element_blank()
  ))
```

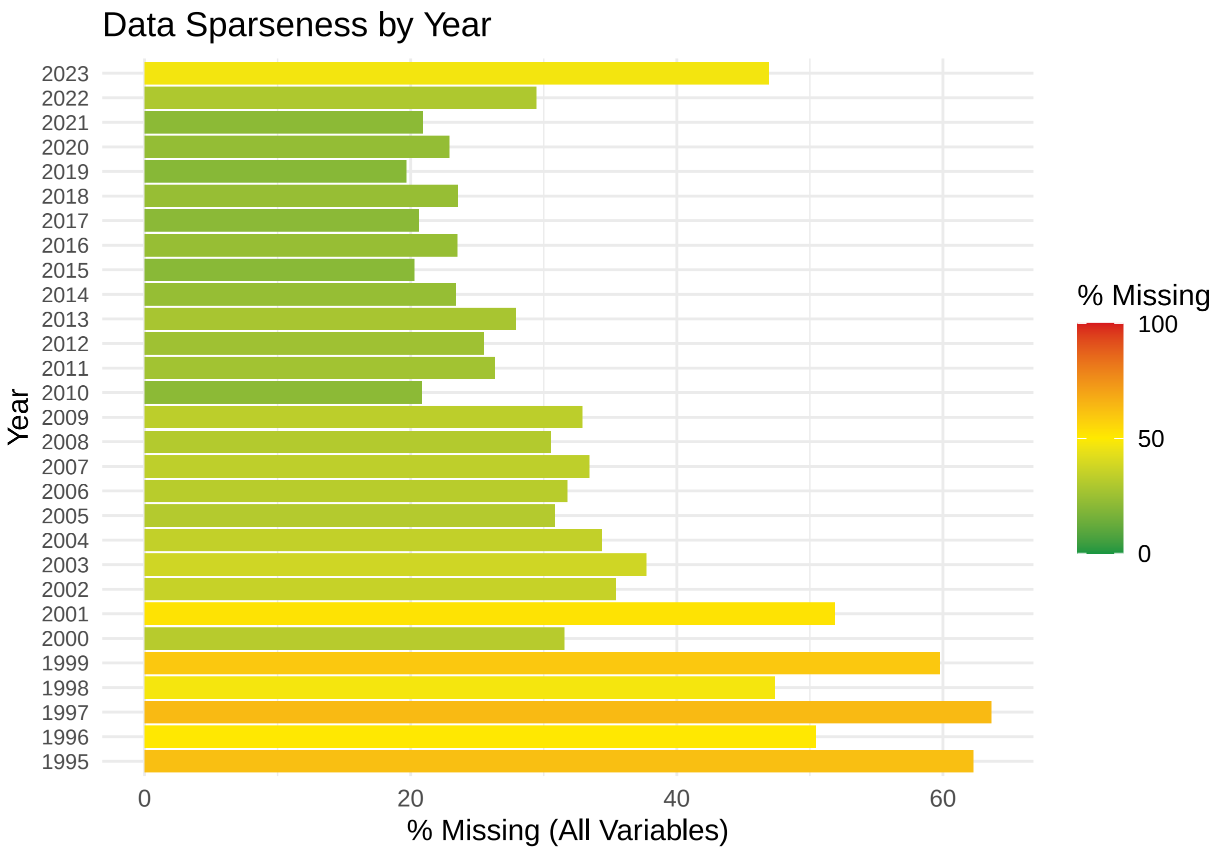



```
FirstCutData %>%
  summarise(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(cols = everything(), names_to = "Variable", values_to = "MissingPct") %>%
  ggplot(aes(x = reorder(Variable, MissingPct), y = MissingPct, fill = MissingPct)) +
  geom_col() +
  coord_flip() +
  scale_fill_gradientn(
    colors = stoplight,
    limits = c(0, 100),
```

```r
    breaks = c(0, 25, 50, 75, 100),  # include 0 and 100
    labels = c("0", "25", "50", "75", "100"),
    name = "% Missing"
  ) +
  labs(
    title = "Overall Sparseness by Variable",
    x = "Variable",
    y = "% Missing"
  ) +
  theme_minimal()
```



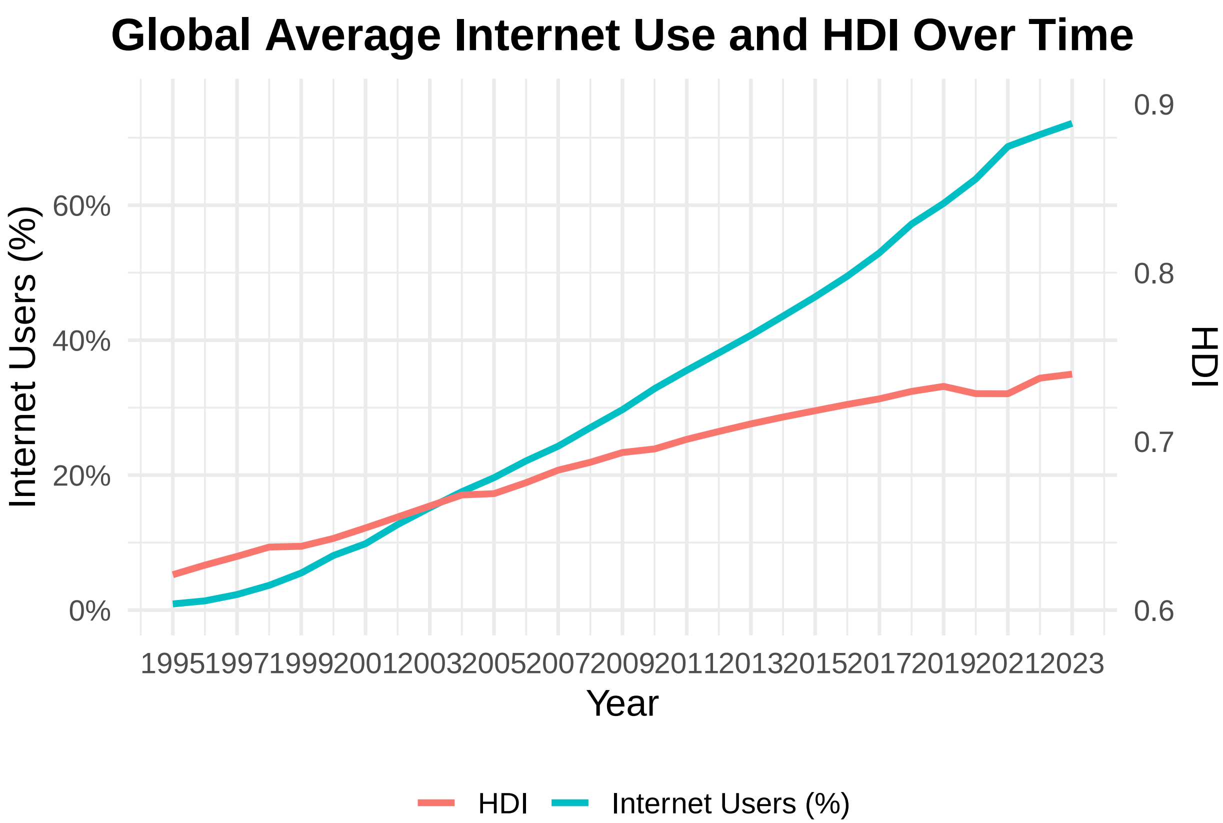
```r
FirstCutData %>%
  group_by(year) %>%
  summarise(across(everything(), ~ mean(is.na(.)) * 100)) %>%
  pivot_longer(-year, names_to = "variable", values_to = "pct_missing") %>%
  group_by(year) %>%
  summarise(pct_missing = mean(pct_missing)) %>%
  arrange(year) %>%
  ggplot(aes(y = factor(year), x = pct_missing, fill = pct_missing)) +
  geom_col() +
  scale_fill_gradientn(
    colors = stoplight,
    values = rescale(c(0, 50, 100)),
    name = "% Missing",
    limits = c(0, 100),
    breaks = c(0, 50, 100),
    labels = c("0", "50", "100")
  ) +
  labs(
    y = "Year",
    x = "% Missing (All Variables)",
    title = "Data Sparseness by Year"
  ) +
  theme_minimal() +
  theme(
    axis.text.y = element_text(size = 8),
    legend.position = "right"
  )
```




```
# Time series analysis

# Overall averages





# Prep data
FirstCut_Yearly_Avg <- FirstCutData %>%
  group_by(year) %>%
  summarize(
```

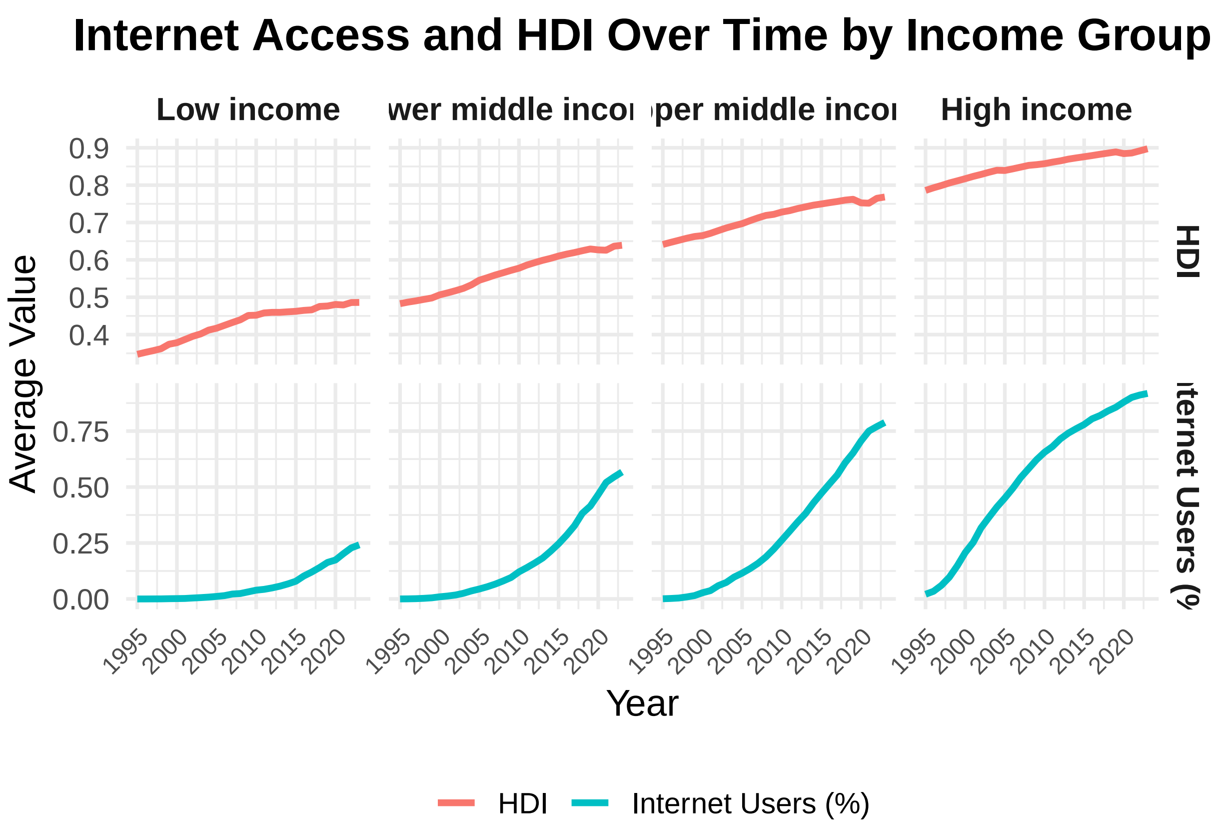
```r
    InternetUsers = mean(InternetUsersPct, na.rm = TRUE)/100,
    HDI = mean(HDI_Index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    HDI_scaled = (HDI - 0.6) / 0.4  # rescale onto same scale as inernet users
  )

# Plot with dual axes
ggplot(FirstCut_Yearly_Avg, aes(x = year)) +
  geom_line(aes(y = InternetUsers, color = "Internet Users (%)"), size = 1.2) +
  geom_line(aes(y = HDI_scaled, color = "HDI"), size = 1.2) +
  scale_x_continuous(breaks = seq(1995, 2025, by = 2))+
  scale_y_continuous(
    name = "Internet Users (%)",
    labels = scales::percent,
    limits = c(0, 0.75),  # force axis to run from 0% to 75%
    sec.axis = sec_axis(~ . * 0.4 + 0.6, name = "HDI") # label axis correctly
  ) +
  scale_color_manual(values = c("Internet Users (%)" = "#00BFC4", "HDI" = "#F8766D")) +
  labs(
    title = "Global Average Internet Use and HDI Over Time",
    x = "Year",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    plot.title = element_text(face = "bold", hjust = 0.5)
  )
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```



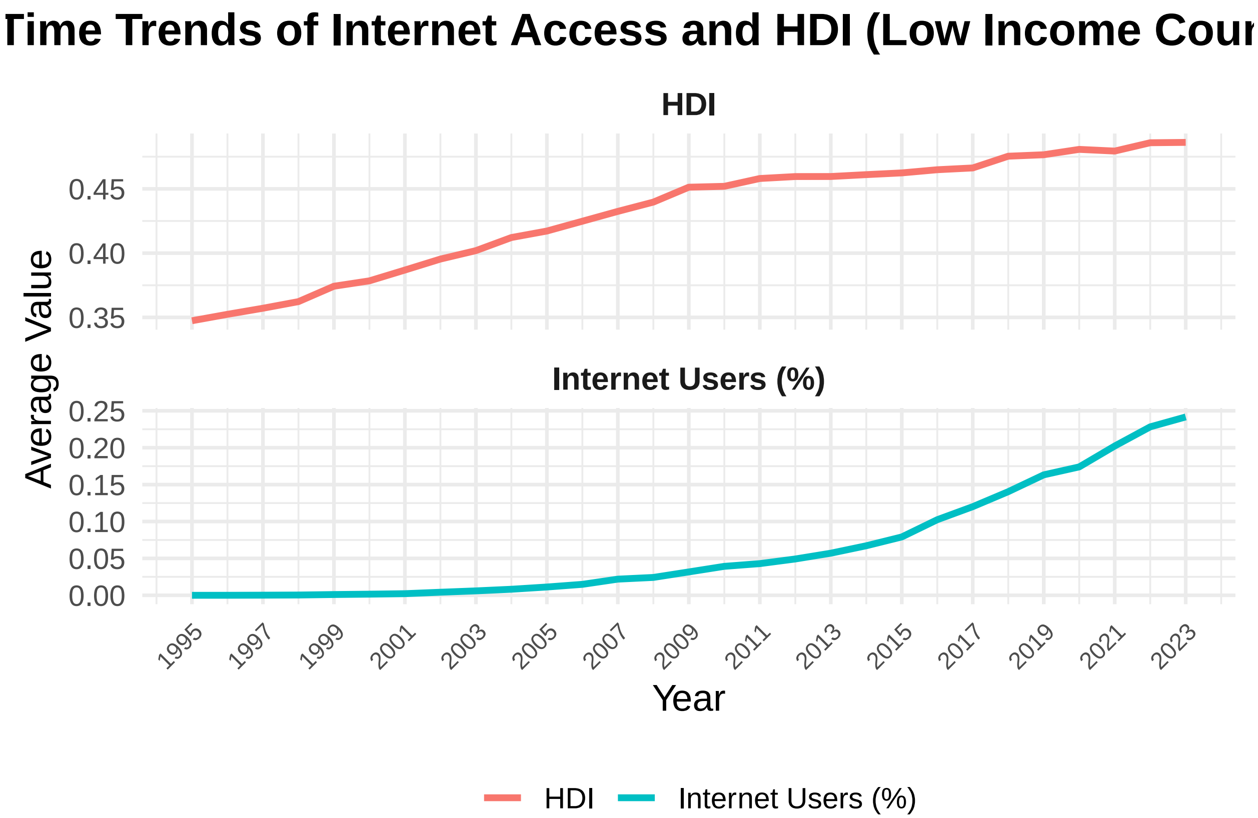

```r
# Group and summarize
FirstCut_Yearly_Income_Avg <- FirstCutDataFiltered %>%
  group_by(year, `Income Group`) %>%
  summarize(
    `Internet Users (%)` = mean(InternetUsersPct, na.rm = TRUE)/100,
    `HDI` = mean(HDI_Index, na.rm = TRUE),
    .groups = "drop"
  )

# Pivot to long format
FirstCut_Yearly_Income_Avg_Long <- FirstCut_Yearly_Income_Avg %>%
  pivot_longer(cols = c(`Internet Users (%)`, `HDI`),
               names_to = "Metric", values_to = "Value")
```

```r
# Plot - facet by Income Group (columns) and Metric (rows), color by Metric
ggplot(FirstCut_Yearly_Income_Avg_Long,
       aes(x = year, y = Value, color = Metric)) +
  geom_line(size = 1.2) +
  facet_grid(Metric ~ `Income Group`, scales = "free_y") +
  scale_color_manual(values = c(
    "HDI" = "#F8766D",
    "Internet Users (%)" = "#00BFC4"
  )) +
  labs(
    title = "Internet Access and HDI Over Time by Income Group",
    x = "Year",
    y = "Average Value",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(size = 9, angle = 45, hjust = 1)  #   the fix
  )
```




```
# Filter to Low Income only
LowIncome_Yearly_Avg <- FirstCutDataFiltered %>%
  filter(`Income Group` == "Low income") %>%
  group_by(year) %>%
  summarize(
    `Internet Users (%)` = mean(InternetUsersPct, na.rm = TRUE)/100,
    `HDI` = mean(HDI_Index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(cols = c(`Internet Users (%)`, `HDI`),
               names_to = "Metric", values_to = "Value")

# Plot with HDI and Internet Users (%) stacked vertically
```

```r
ggplot(LowIncome_Yearly_Avg, aes(x = year, y = Value, color = Metric)) +
  geom_line(size = 1.2) +
  facet_wrap(~ Metric, ncol = 1, scales = "free_y") +
  scale_color_manual(values = c(
    "HDI" = "#F8766D",
    "Internet Users (%)" = "#00BFC4"
  )) +
  scale_x_continuous(breaks = seq(1995, 2025, 2)) +
  labs(
    title = "Time Trends of Internet Access and HDI (Low Income Countries)",
    x = "Year",
    y = "Average Value",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(size = 9, angle = 45, hjust = 1)
  )
```




```
# Create groupings
LowIncome_Yearly_Avg <- FirstCutDataFiltered %>%
  filter(`Income Group` == "Low income") %>%
  group_by(year) %>%
  summarize(
    `Internet Users (%)` = mean(InternetUsersPct, na.rm = TRUE)/100,
    `HDI` = mean(HDI_Index, na.rm = TRUE),
    `IHDI` = mean(IHDI_Index, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  pivot_longer(cols = c(`Internet Users (%)`, `HDI`, `IHDI`),
               names_to = "Metric", values_to = "Value") %>%
  mutate(
```

```r
    MetricGroup = ifelse(Metric %in% c("HDI", "IHDI"),
                         "Wellbeing Index",
                         "Internet Access")
  )

# Reorder facet levels so Wellbeing is on top
LowIncome_Yearly_Avg$MetricGroup <- factor(
  LowIncome_Yearly_Avg$MetricGroup,
  levels = c("Wellbeing Index", "Internet Access")
)

# Plot
ggplot(LowIncome_Yearly_Avg, aes(x = year, y = Value, color = Metric)) +
  geom_line(size = 1.2) +
  facet_wrap(~ MetricGroup, ncol = 1, scales = "free_y") +
  scale_color_manual(values = c(
    "HDI" = "#F8766D",
    "IHDI" = "#D89000",
    "Internet Users (%)" = "#00BFC4"
  )) +
  scale_x_continuous(breaks = seq(1995, 2025, 2)) +
  labs(
    title = "HDI, IHDI, and Internet Access Over Time (Low Income Countries)",
    x = "Year",
    y = "Average Value",
    color = ""
  ) +
  theme_minimal(base_size = 14) +
  theme(
    legend.position = "bottom",
    strip.text = element_text(face = "bold", size = 12),
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(size = 9, angle = 45, hjust = 1)
  )
```

```
## Warning: Removed 15 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

IDI, IHDI, and Internet Access Over Time (Low Income Cou